



University of Stuttgart
Germany

Mathematical modeling and simulation
of cellular systems (ISA)

Automated Design and Analysis of Gene Regulatory Networks for Simulating Complex Spatial Patterns

Master's Thesis
Loghman Samani
February 2025

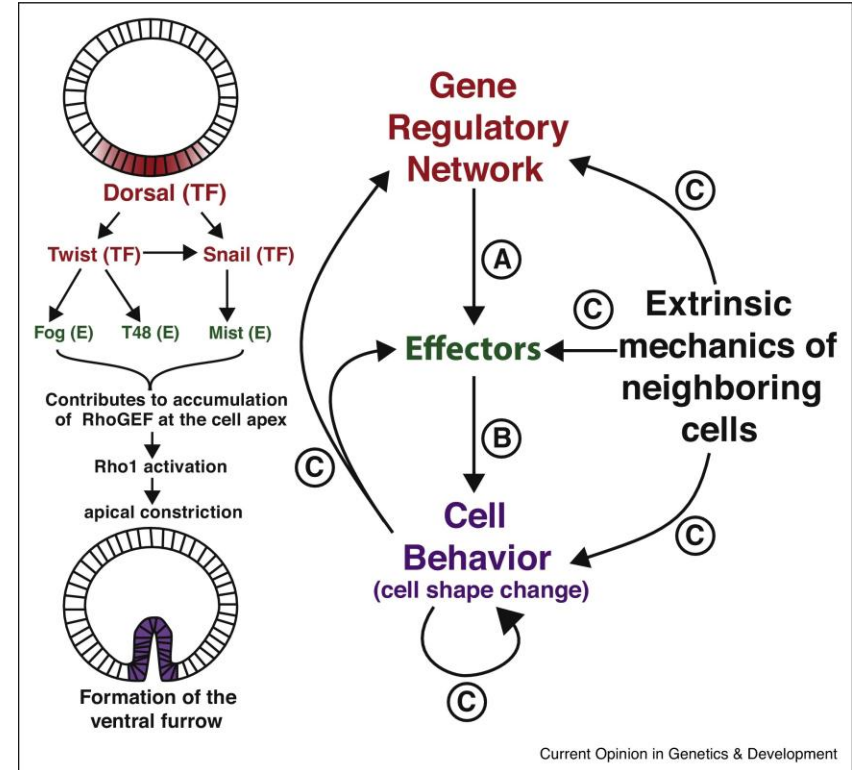
Gene Regulatory Networks (GRNs)

Definition

- Networks of interacting genes, proteins, and molecules regulating gene expression

Importance

- Control of cell processes (e.g., division, differentiation)
- Response to environmental changes.
- Disease Understanding: Key in cancer, neurodegenerative diseases
- Synthetic Biology: Enables creation of custom cellular functions



*The influence of GRNs on tissue morphogenesis
Smith et al. (2018)*

“Automatic Design of Gene Regulatory Mechanisms for Spatial Pattern Formation”

- Mossavi and Lobo (2024)

Inputs

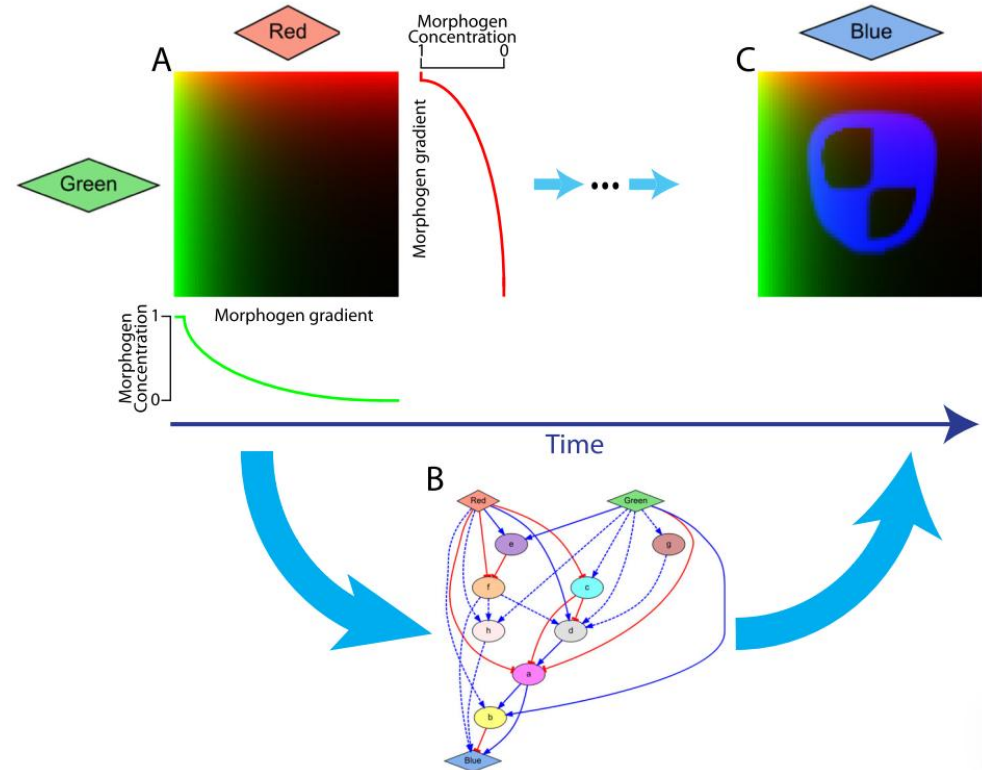
- Two orthogonal **static gradients** released from the edges (top and left) in a 2D cell culture.

Mechanism

- Creating a Gene Regulatory Network (GRN) to express a chain of **non-diffusible genes**

Outcome

- **Reporter gene** expression forms a stable spatial pattern matching a target pattern

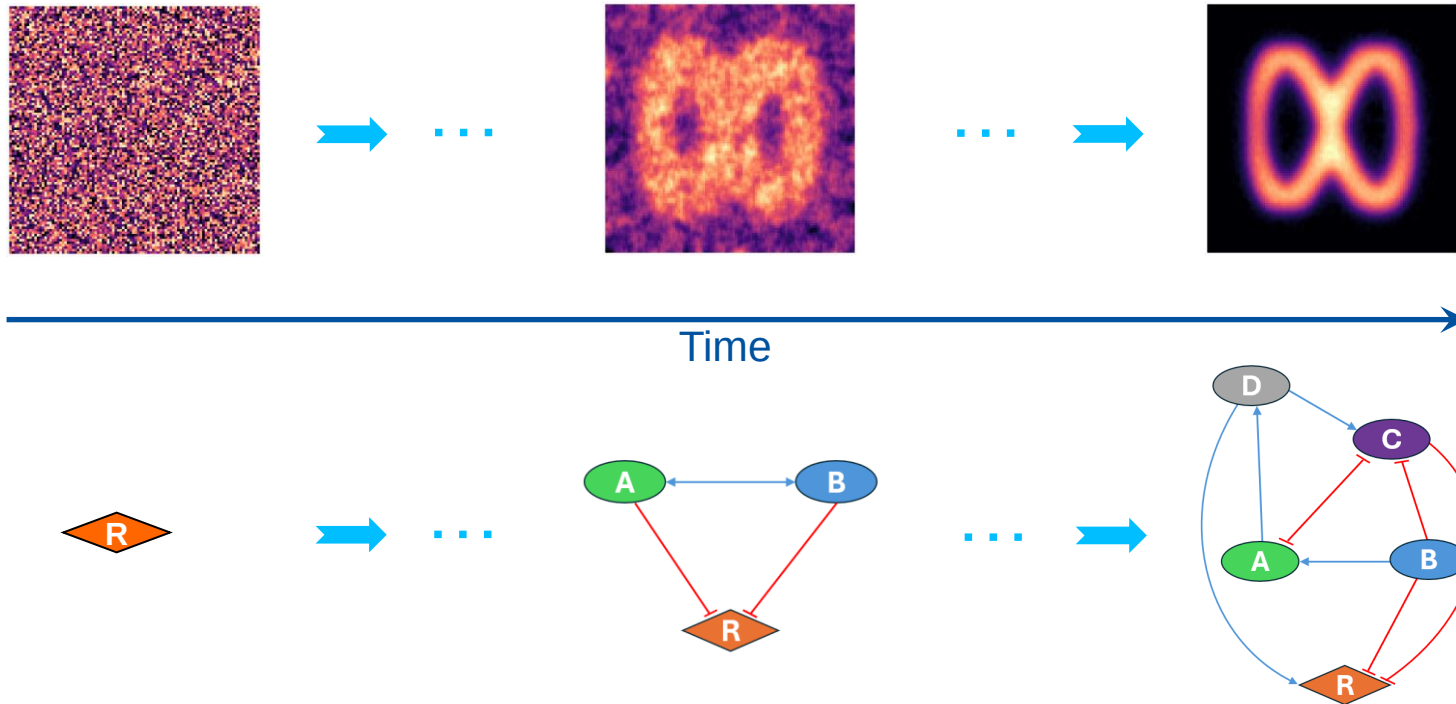


*Designing Gene Regulatory Networks
Mousavi & Lobo (2024)*

Aim of the Study

Goal

- Construct and optimize a Gene Regulatory Network (GRN) to produce a **diffusible reporter gene** product that matches a target spatial pattern



Aim of the Study

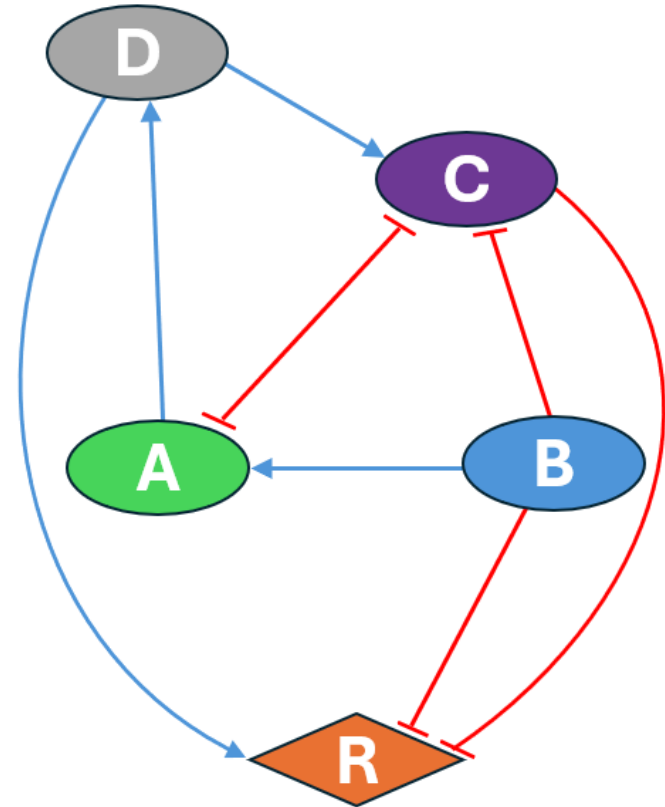
Improvements & Differences

GRN Structure

- Genes and interaction parameters

Diffusible Gene Products

- Influence spatial pattern formation

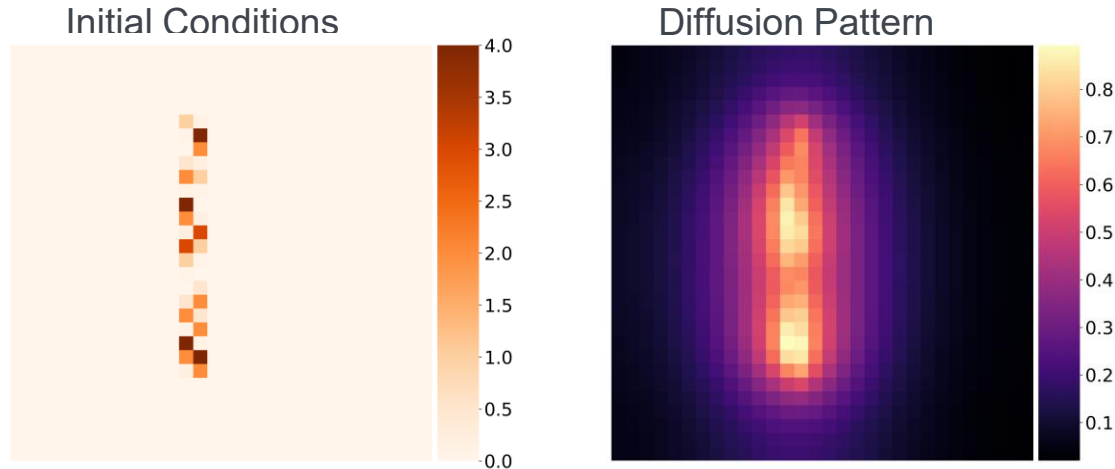


Aim of the Study

Improvements & Differences

Initial Conditions (*Replacing Static Morphogen Gradients*)

- Unique gene expression levels per cell → Non-static gradients
- Shape diffusion & gene product interactions
- Mimic biological variability (stochasticity, positional cues)
- Essential for realistic GRN simulations

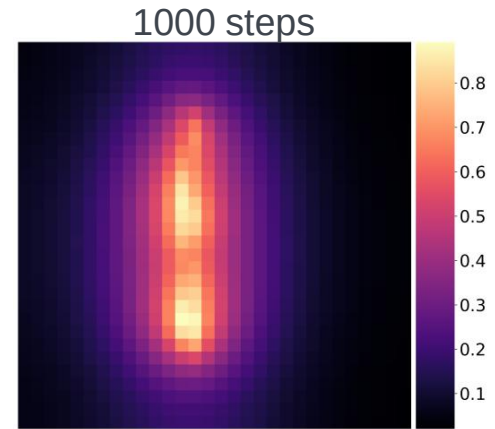
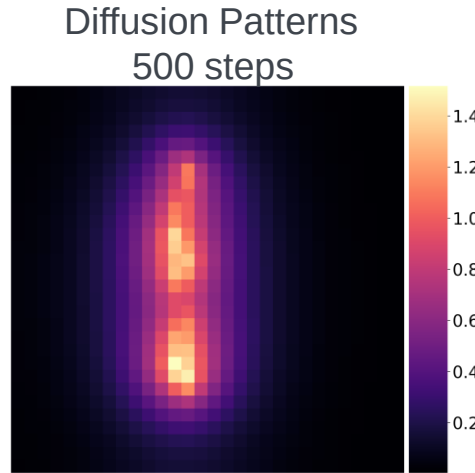
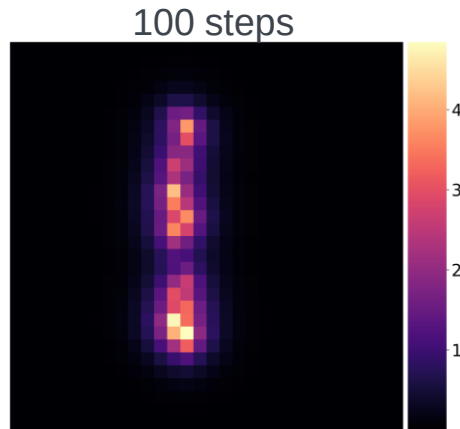
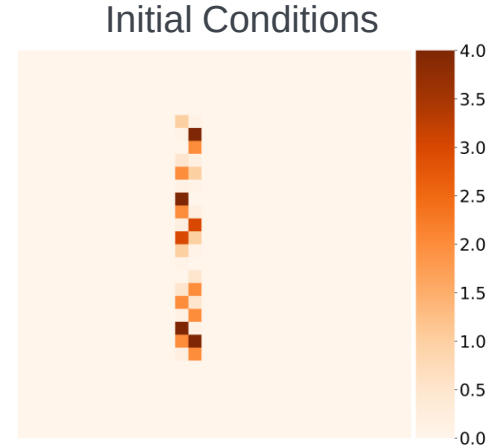


Aim of the Study

Improvements & Differences

Simulation Settings

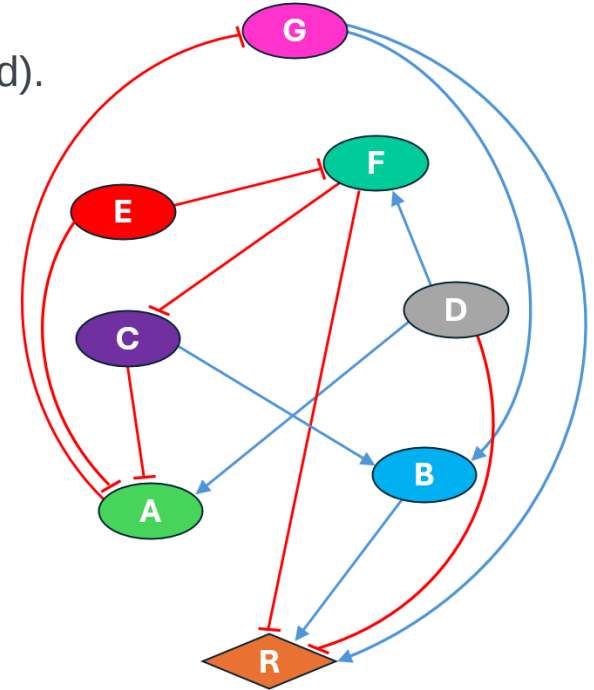
- Duration and time step (step = duration / time Step)



Methods

Mossavi & Lobo (2024) Approach

- Used a genetic algorithm (part of the evolutionary algorithms family)
- Evolutionary algorithms are suitable for
 - Large search spaces (when prior information is limited).
 - Low parameter counts (~100 or fewer)



Methods

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Same GRN, Different Complexity

- 8 genes and 13 interactions

Their Case

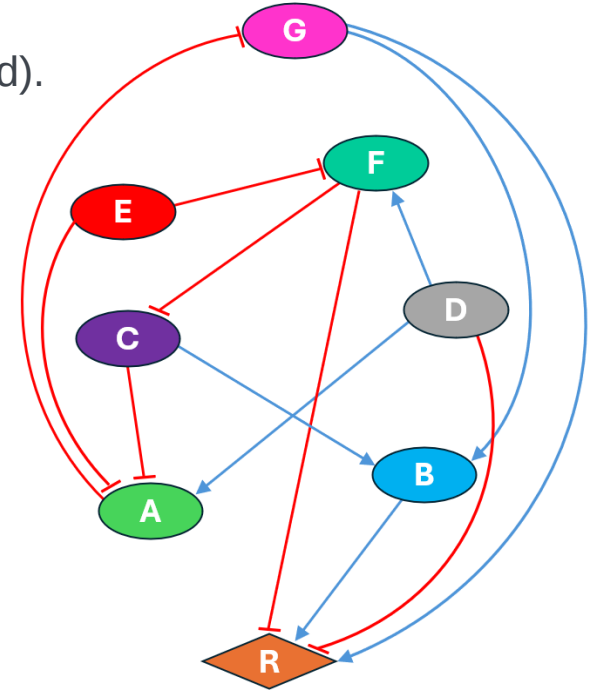
- Total Parameters: 42

Our Case

- Total Parameters: 80,063

Conclusion

- A genetic algorithm alone is not sufficient due to the increased system complexity and parameter count



Methods

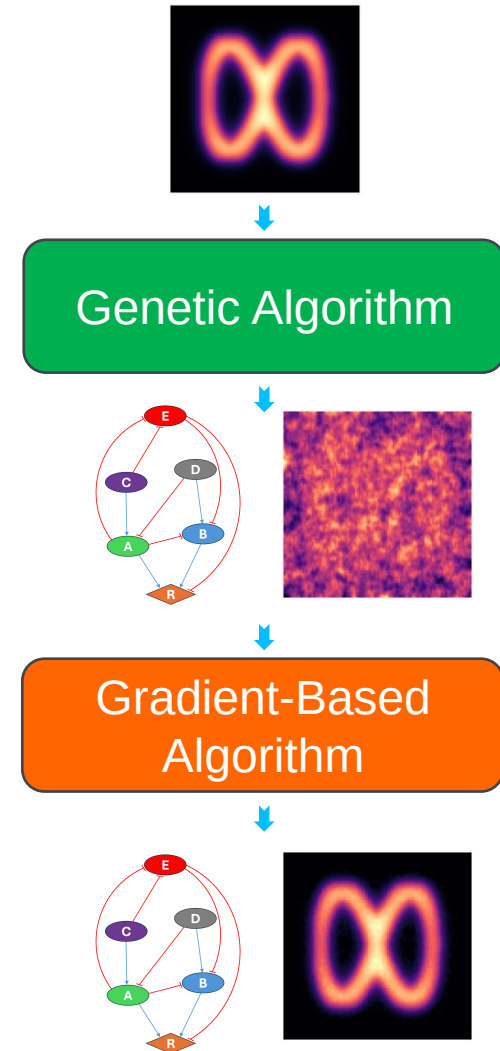
Hybrid Optimization Approach

Genetic Algorithm

- Constructs GRN Structure
- Preliminary Convergence to Global Minimum by optimizing
 - Gene and interaction rate constants
 - Initial conditions
 - Simulation settings (duration, time step)

Gradient-Based Optimization

- Fine-tunes the System by optimizing
 - Gene and interaction rate constants
 - Initial conditions



Methods

Genetic Algorithm Structure

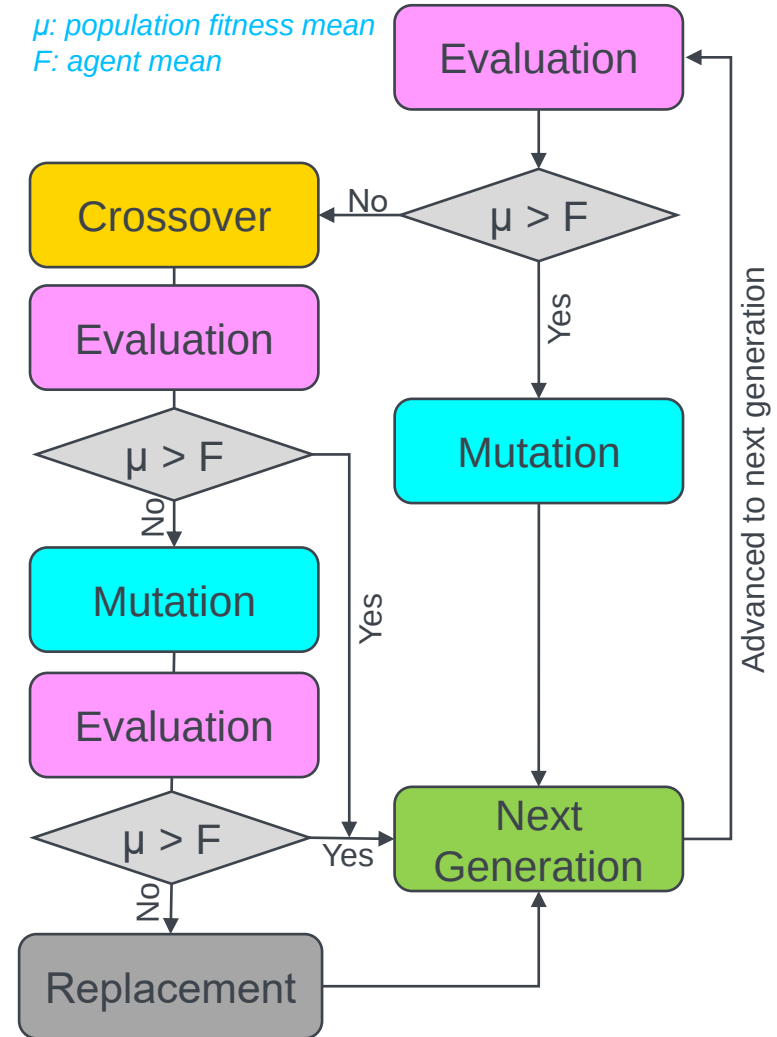
- “Genetic Algorithm Based on Natural Selection Theory for Optimization Problems”
 - Albadr et al., (2020)

Key Processes

- Evaluation
- Selection
- Crossover
- Mutation

Population-Based Optimization

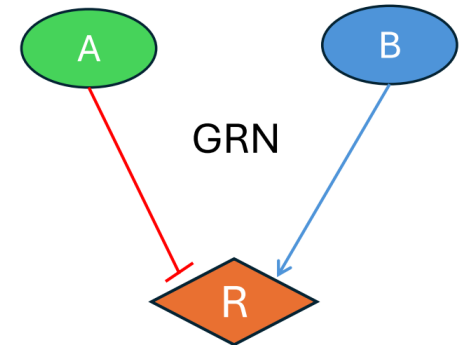
- Operates on a population of potential solutions and evolves over generations



Methods

Agent Structure

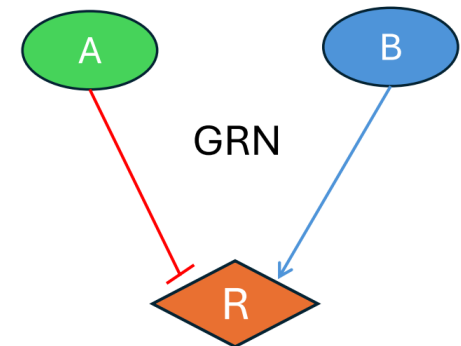
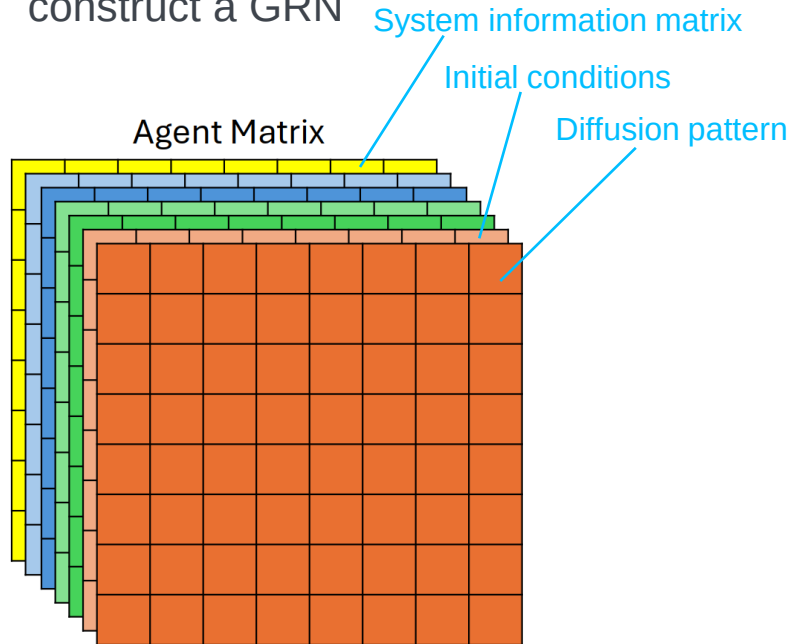
- An agent represents a potential solution for optimization. In our case, each solution is a Gene Regulatory Network (GRN).



Methods

Agent Structure

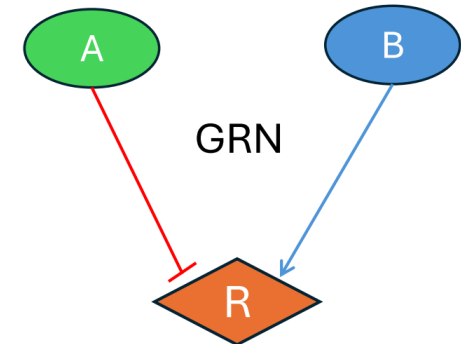
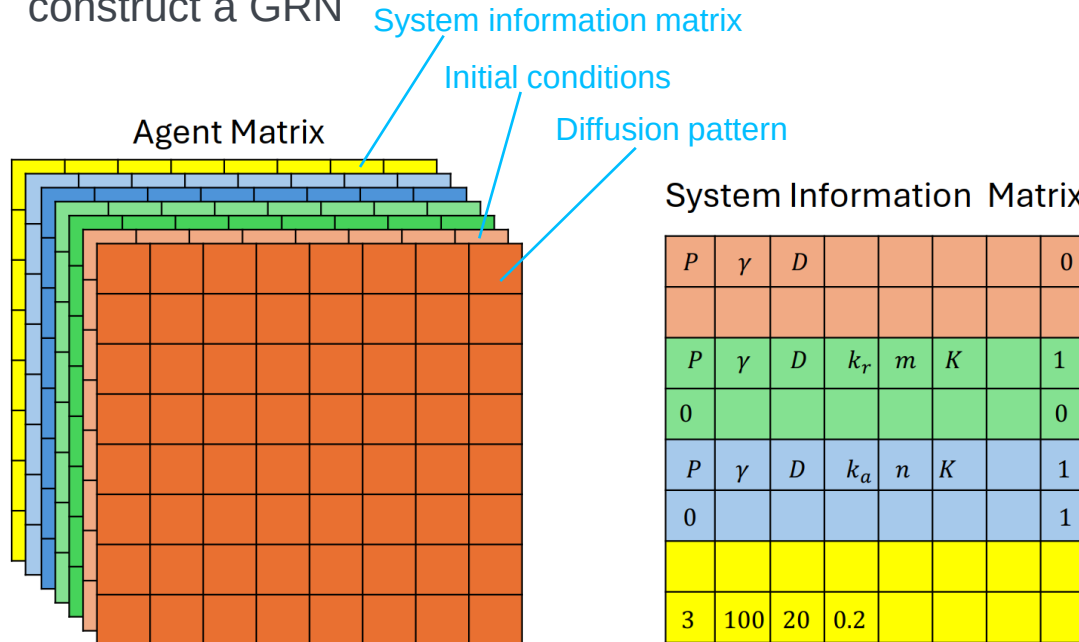
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- An agent is defined as a 3D matrix containing all necessary information to construct a GRN



Methods

Agent Structure

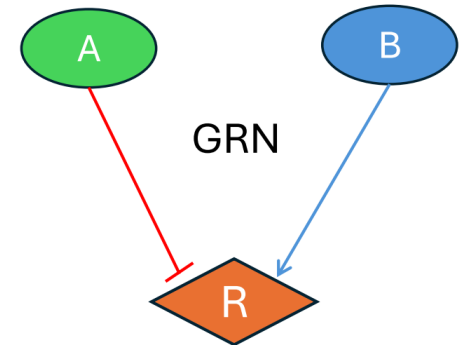
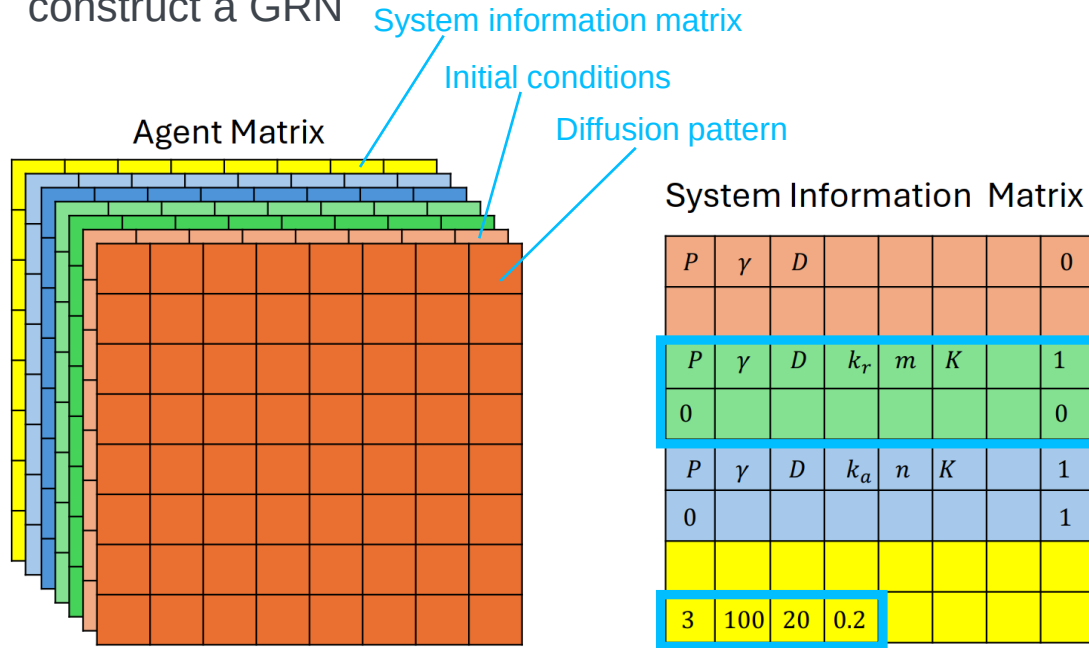
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Methods

Evaluation: Simulation

- Models molecular interactions in a 2D environment.
- Governed by nonlinear **PDEs** with spatial diffusion
- Gene regulation follows Hill equations
- No-flux (Neumann) boundary conditions

$$\frac{\partial g_i}{\partial t} = F_i(\{g_j\}) + D_i \nabla^2 g_i$$
$$F_i(\{g_j\}) = P_i \rho_i(x, y) + \left(\sum_{j \in A_i} k_{a_{ij}} \frac{g_j^{n_{ij}}}{K_{ij}^{n_{ij}} + g_j^{n_{ij}}} - \sum_{k \in R_i} k_{r_{ik}} \frac{g_k^{m_{ik}}}{K_{ik}^{m_{ik}} + g_k^{m_{ik}}} \right) - \gamma_i g_i$$

$\frac{\partial g_i}{\partial t}$: gene expression dynamics

$D_i \nabla^2 g_i$: diffusion term

$F_i(\{g_i\})$: regulatory term (net production)

$\gamma_i g_i$: decay term

$\rho_i(x, y)$: initial conditions

Methods

Evaluation: Objective Function

- Compares GRN simulation results with the target pattern
- The goal is to minimize the objective function

$$\text{MIN}[L_t(\hat{y}, y) = \alpha L_{mse}(\hat{y}, y) + \beta L_{ssim}(\hat{y}, y)]$$

$$L_{mse}(\hat{y}, y) = \frac{1}{wh} \sum_{i=1}^w \sum_{j=1}^h \left(\hat{y}_{(i,j)} - y_{(i,j)} \right)^2$$

Objective Function Components

- Mean Squared Error (MSE)
 - Measures accuracy
- Structural Similarity Index (SSIM) .
 - Captures structural alignment

$$L_{ssim}(\hat{y}, y) = 1 - \frac{(2\mu_{\hat{y}}\mu_y + C_1)(2\sigma_{\hat{y}y} + C_2)}{(\mu_{\hat{y}}^2 + \mu_y^2 + C_1)(\sigma_{\hat{y}}^2 + \sigma_y^2 + C_2)}$$

$\hat{y}, y$: constructed, target pattern

$\mu_{\hat{y}}, \mu_y$: mean of constructed, target pattern

$\sigma_{\hat{y}}^2, \sigma_y^2$: variance of constructed, target pattern

$\sigma_{\hat{y}y}$: covariance of $\hat{y}$ and y

C_x : small number to avoid zero division

Methods

Mutation

- Introduces diversity through random changes based on a mutation rate
- Uses **Differential Variation Mutation**

Six Mutation Processes

- Simulation Settings Mutation (duration, time step)
- Initial Conditions Mutation
- Rate Constants Mutation
- Gene Insertion Mutation
- Interaction Insertion Mutation
- Interaction Deletion Mutation

$$\hat{p} = p + F(p_{e2} - p_{e1})$$

$\hat{p}$: mutated parameter

p : original parameter

*$p_{e1,e2}$: Same parameter chosen from two
randomly selected elite agents*

F : Mutation constant, set to 0.8

Elite agent: An agent with a better solution

Methods

Crossover

- Encourages similarity by recombining agent parameters.

Method: Arithmetic Crossover

- Combines parameters from two parent agents to generate offspring

Crossover Applies To

- System Settings (duration, time step)
- Initial Conditions
- Rate Constants

$$\hat{p} = \beta p + (1 - \beta)p_e$$

$$\beta \sim v(-\alpha, 1 + \alpha)$$

$\hat{p}$: updated parameter after crossover

p : original parameter

p_e : same parameter chosen from an elite agent

β : recombination factor

α : crossover hyperparameter (=0.4)

v : uniform distribution

Elite agent: an agent with a better solution

Methods

Gradient-Based Optimization

Idea of Partial Derivative

- Measures how a parameter change affects the objective function
- Represents the **tangent line's slope**

Tangent Line & Optimization Pathway

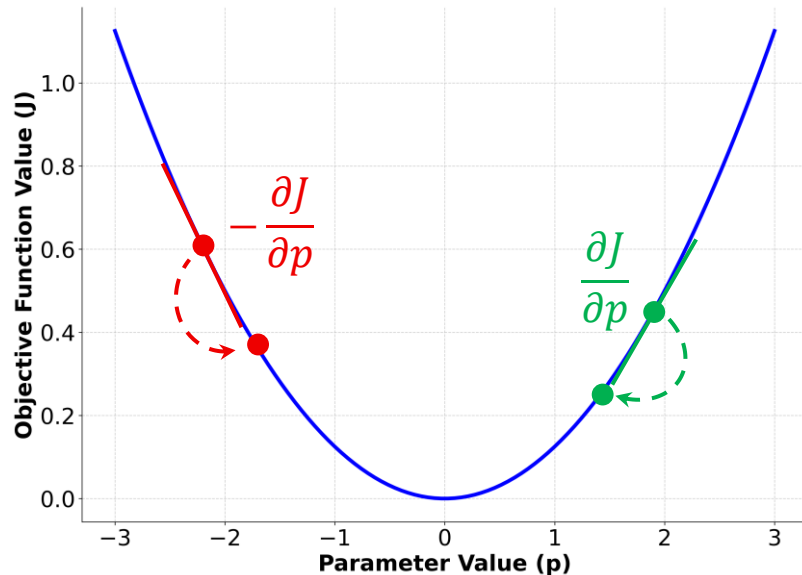
- Shows the direction & rate of change
- Move downhill to minimize the objective function

Learning Rate (α) & Parameter Update

- Controls step size in each iteration

Extending with the Chain Rule

- Gradients propagate through multiple parameters



$$\text{decreased} : p = p - \alpha \frac{\partial J}{\partial p}$$

$$\text{increased} : p = p - \alpha \left(-\frac{\partial J}{\partial p} \right)$$

Methods

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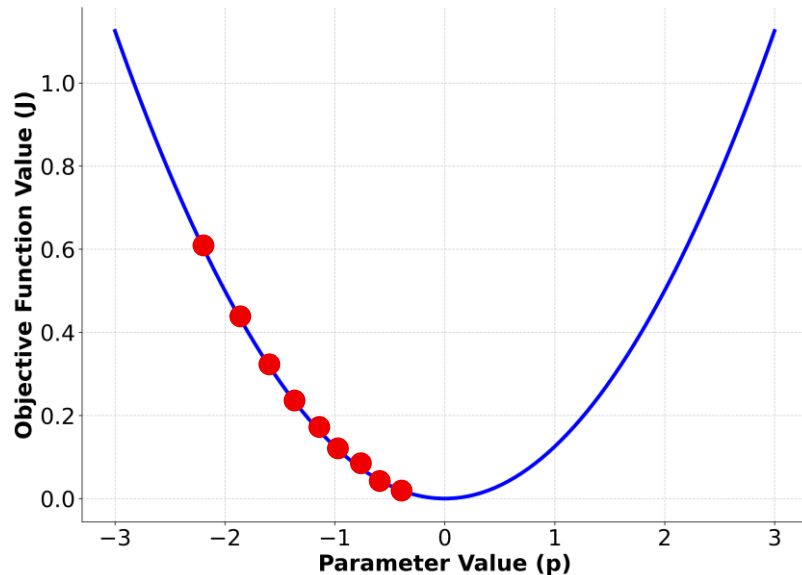
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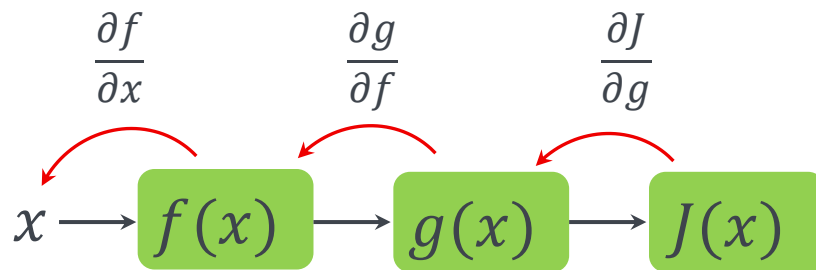
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forward propagation →

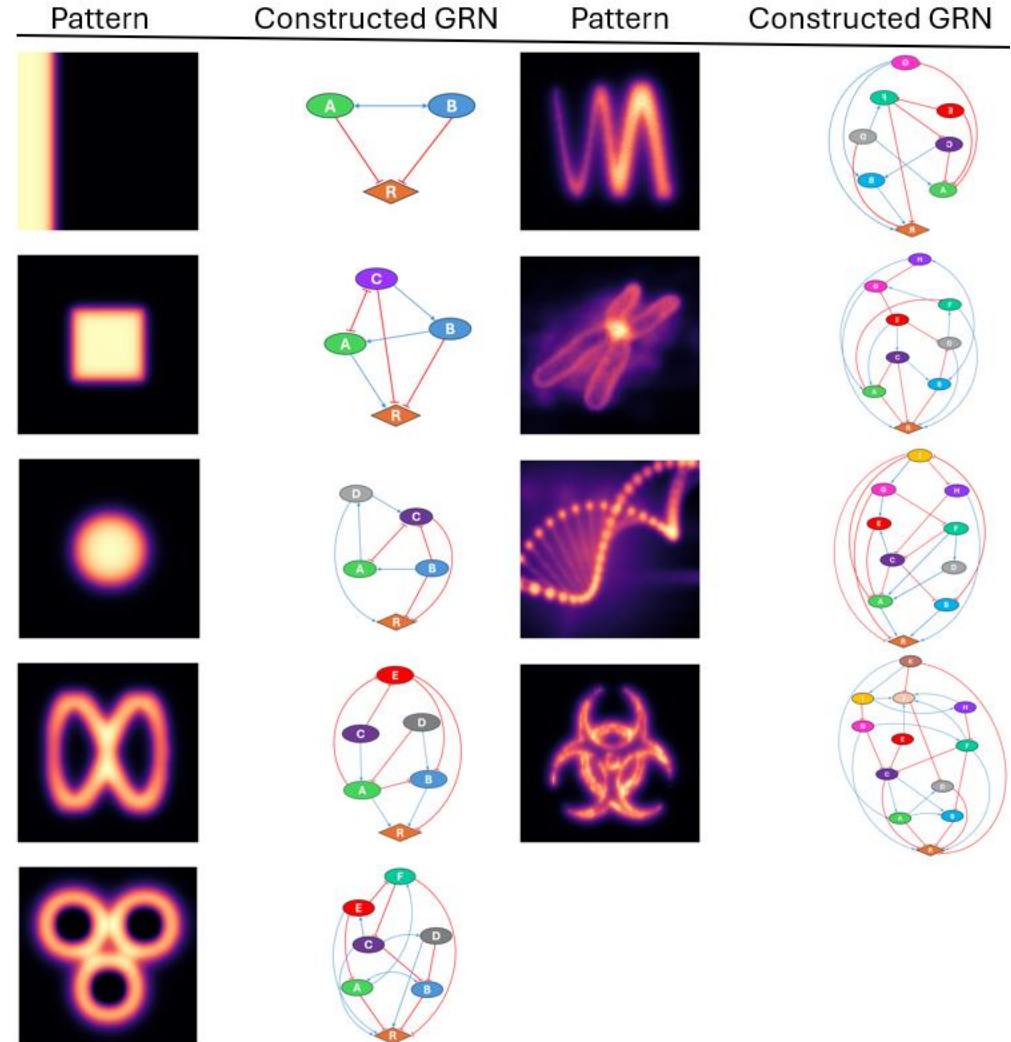


← *backward propagation*

$$\frac{\partial J}{\partial x} = \frac{\partial J}{\partial g} \cdot \frac{\partial g}{\partial f} \cdot \frac{\partial f}{\partial x}$$

Results

- Evaluated the system on 9 pattern generation tasks
- Target patterns: 100×100 pixels
- Simulation run for up to 100 iterations ($\Delta t \approx 0.1$), total time ≈ 10
- Evolutionary optimization: 100 epochs for all cases
- Second stage: Gradient-based optimization for refinement and convergence



Results

Target

- Infinity-Shaped Diffusion Pattern (100x100 pixel figure)

Initial System Configuration

- Single reporter gene
- 200 Agents

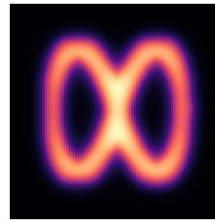
Genetic Algorithm

- Conducted 100 iterations to construct the GRN

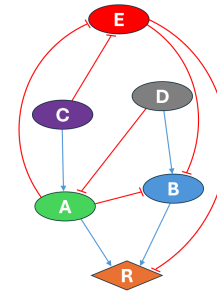
Gradient Descent

- Performed 110 iterations to achieve full convergence of the system

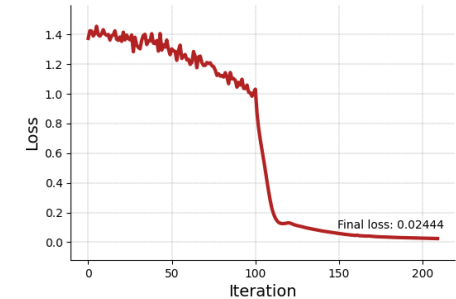
Target Pattern



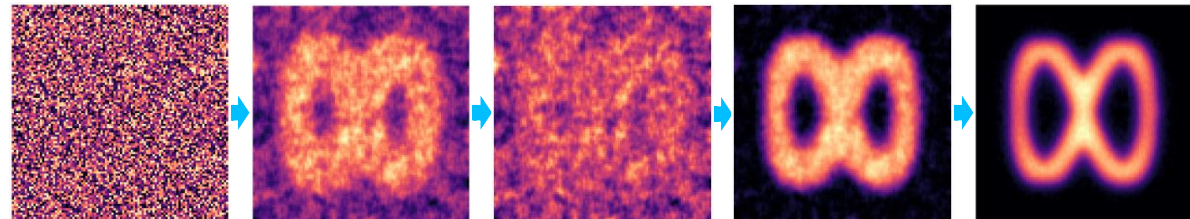
Constructed GRN



Loss vs. Iteration



Optimization Progress →



Constructed Pattern

Results

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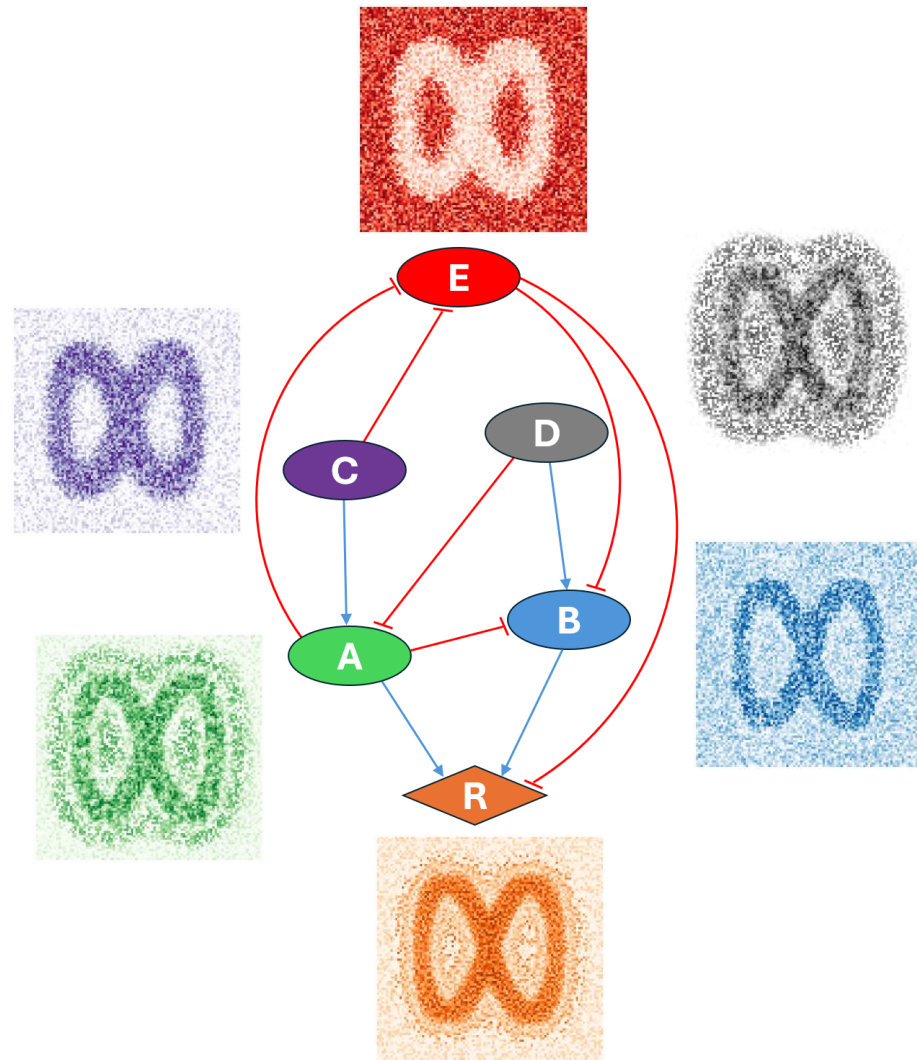
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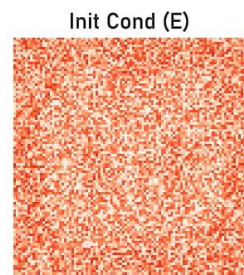
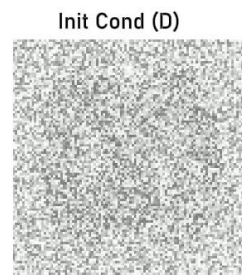
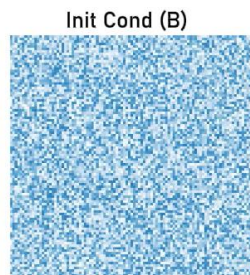
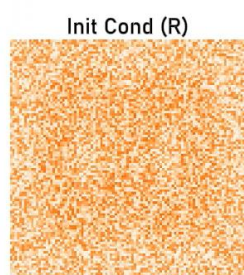
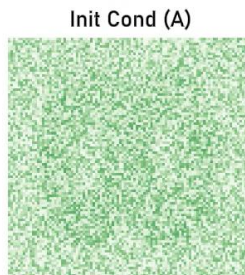
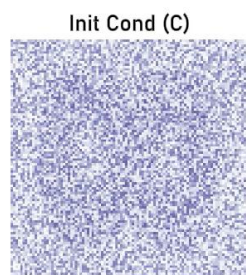
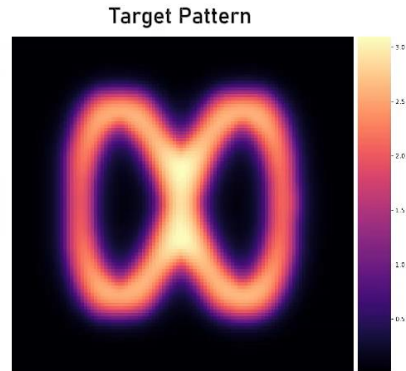
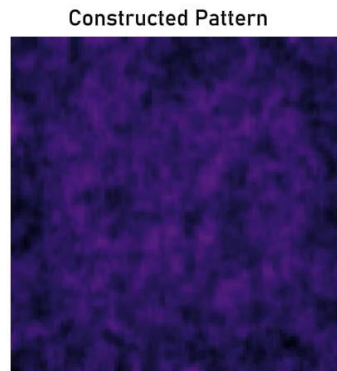
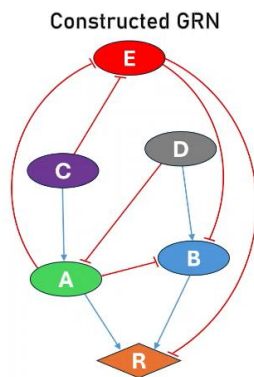


Results

Total Iterations: 210

Optimization Time: 6.5 hr

Final Loss: 0.02444 (α : 0.6, β : 0.4)



Results

Downscaling Effect for Improved System Efficiency

Agent Resolution Reduction

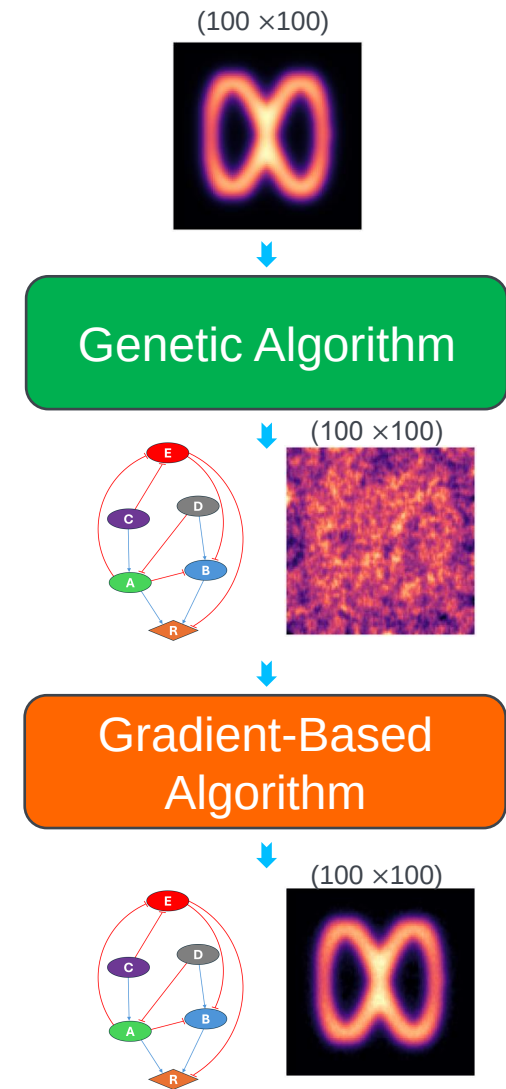
- Lowers the computational cost by decreasing the resolution of agents

Parameter Reduction

- Decreases the number of parameters, effectively reducing the search space.

Enhanced Efficiency

- Improves system performance by lowering computational demands and narrowing the search space, leading to faster optimization



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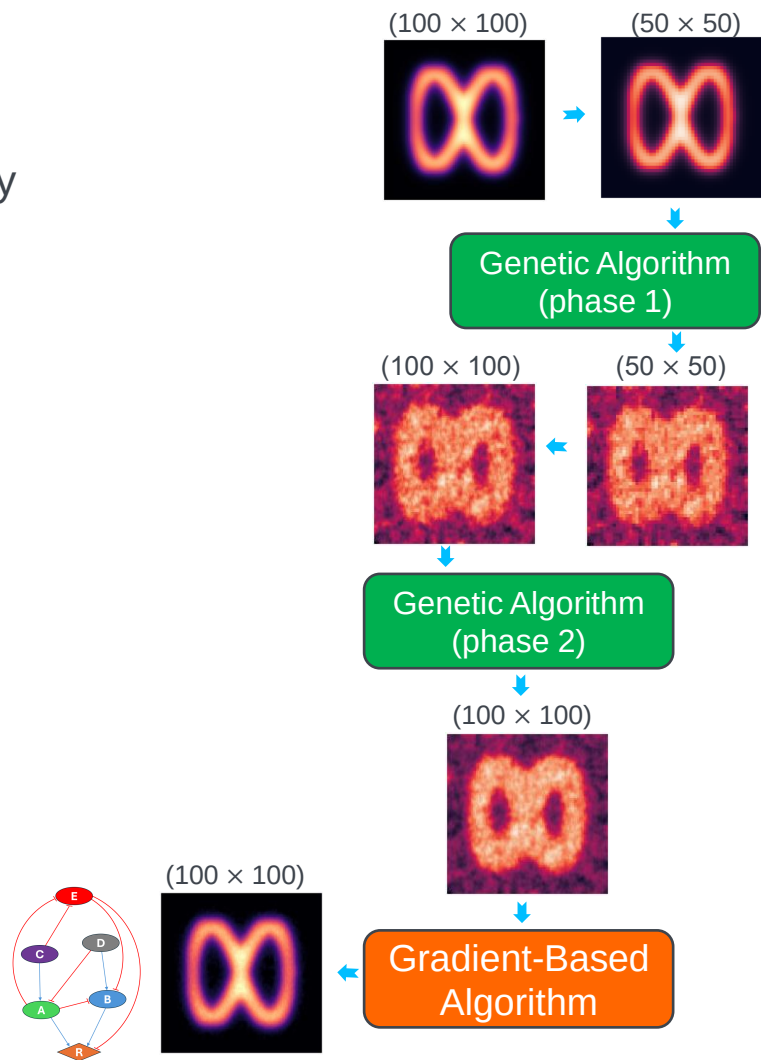
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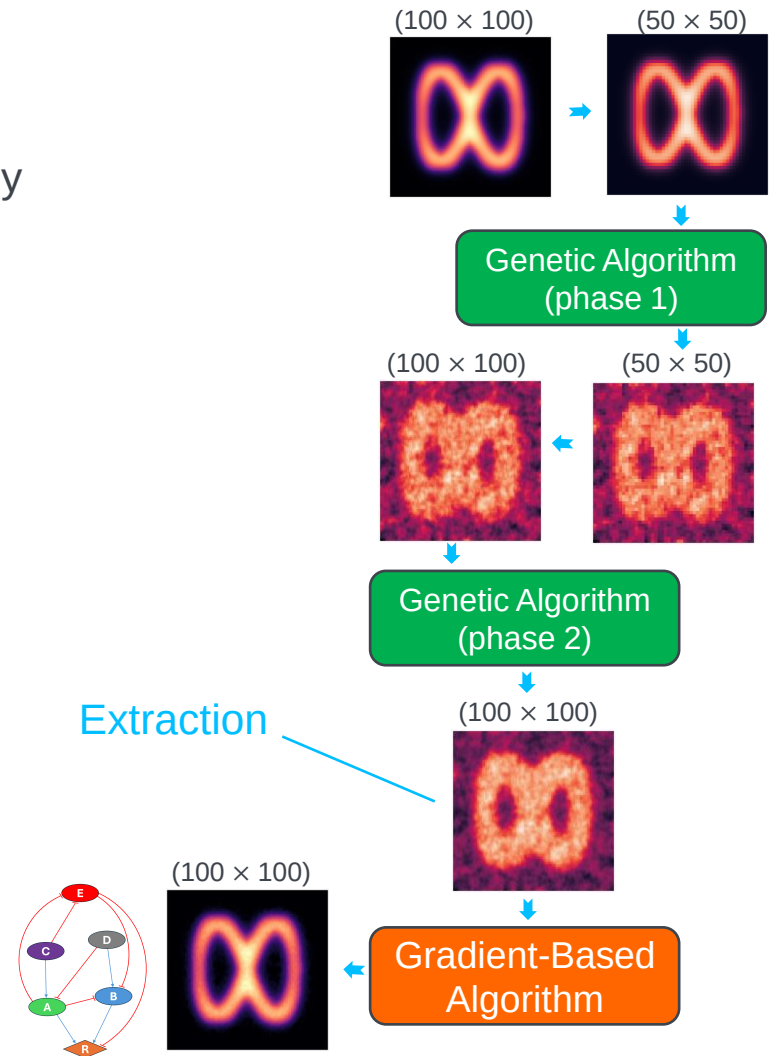
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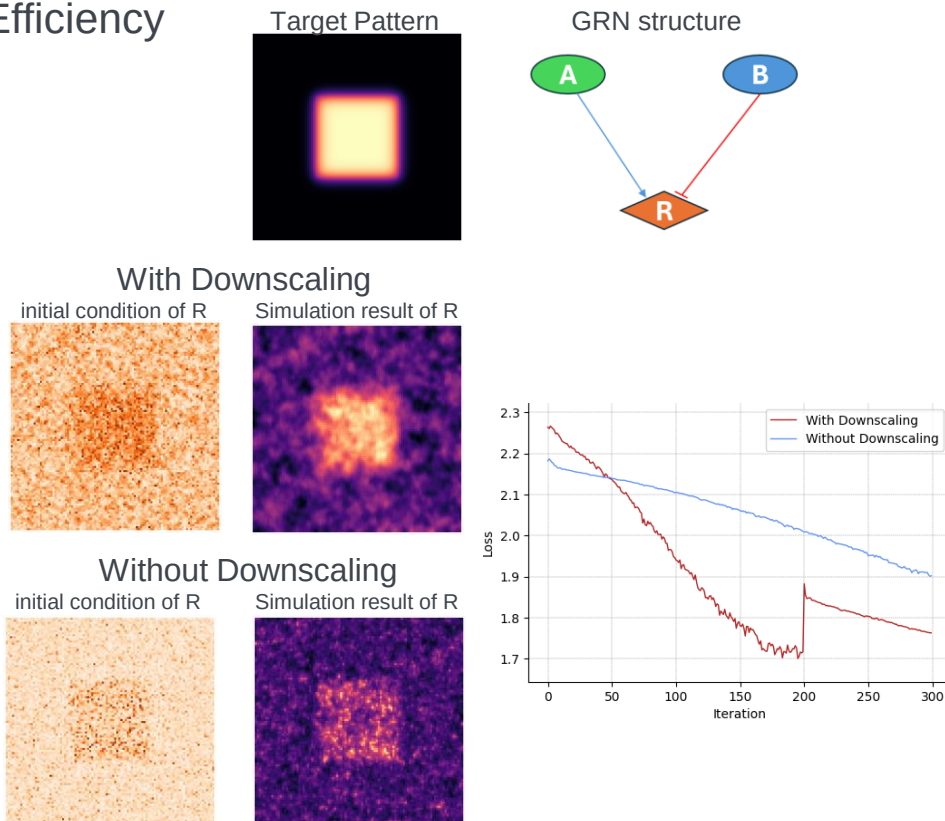
Downscaling Effect for Improved System Efficiency

Three models tested with

- 300 agents
- Fixed GRN
- 300 epochs
 - Phase 1: 200 epochs
 - Phase 2: 100 epochs

Results

- 50% reduction in computation time
- Example: Model 2 took 3,426s with downscaling, 6,332s without
- Faster convergence with downscaling

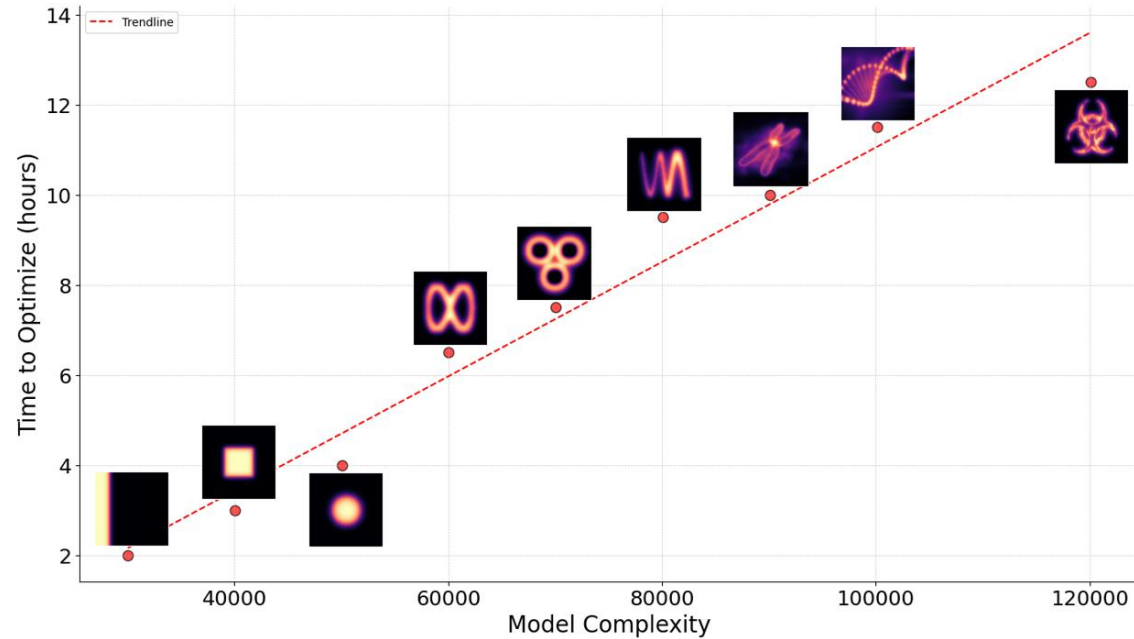


Results

Model Complexity and Optimization Time

Complexity measured by trainable parameters

- Genes
- Initial conditions
- Regulatory interactions



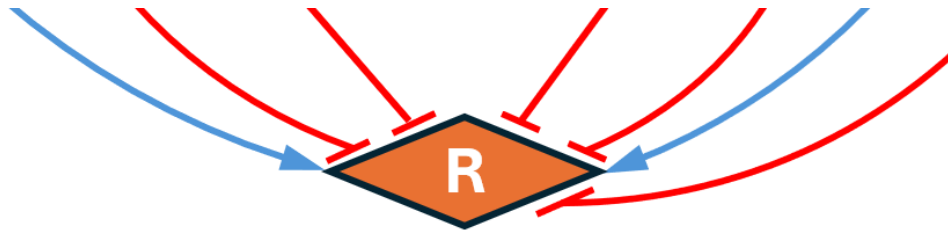
Optimization time increases linearly with complexity

Limitations & Future Work

Simplified Regulatory Model

- Assumes independent & additive interactions
- Incorporate **cooperative & competitive** interactions for more realistic gene regulation

$$F_i(\{g_j\}) = P_i \rho_i(x, y) + \left(\sum_{j \in A_i} k_{a_{ij}} \frac{g_j^{n_{ij}}}{K_{ij}^{n_{ij}} + g_j^{n_{ij}}} - \sum_{k \in R_i} k_{r_{ik}} \frac{g_k^{m_{ik}}}{K_{ik}^{m_{ik}} + g_k^{m_{ik}}} \right) - \gamma_i g_i$$



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Limited Genetic Algorithm Exploration

- Extend GA iterations before fine-tuning to improve global optimization

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Lack of Diversity in Initial Conditions

- Add a diversity term to the objective function to encourage variation

$$\textcolor{blue}{MIN}[L_t(\hat{y}, y) = \alpha L_{mse}(\hat{y}, y) + \beta L_{ssim}(\hat{y}, y) + \gamma \textcolor{green}{div}]$$

$$\textcolor{green}{div} = \sum_{i \neq j} \textcolor{green}{Cov}(c_i, c_j)$$

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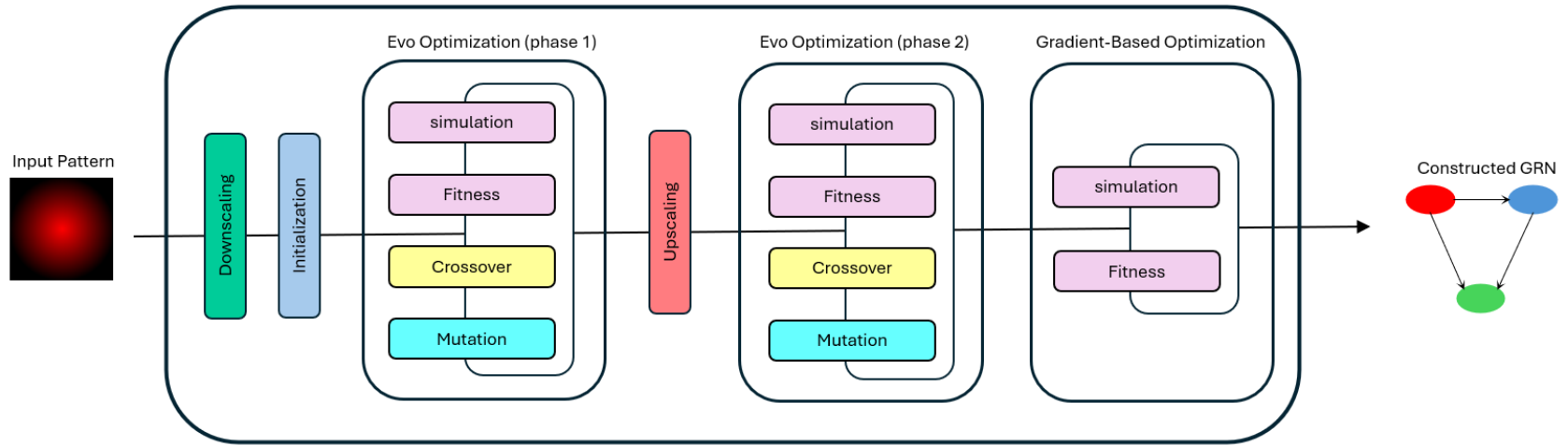
Deterministic Simulations

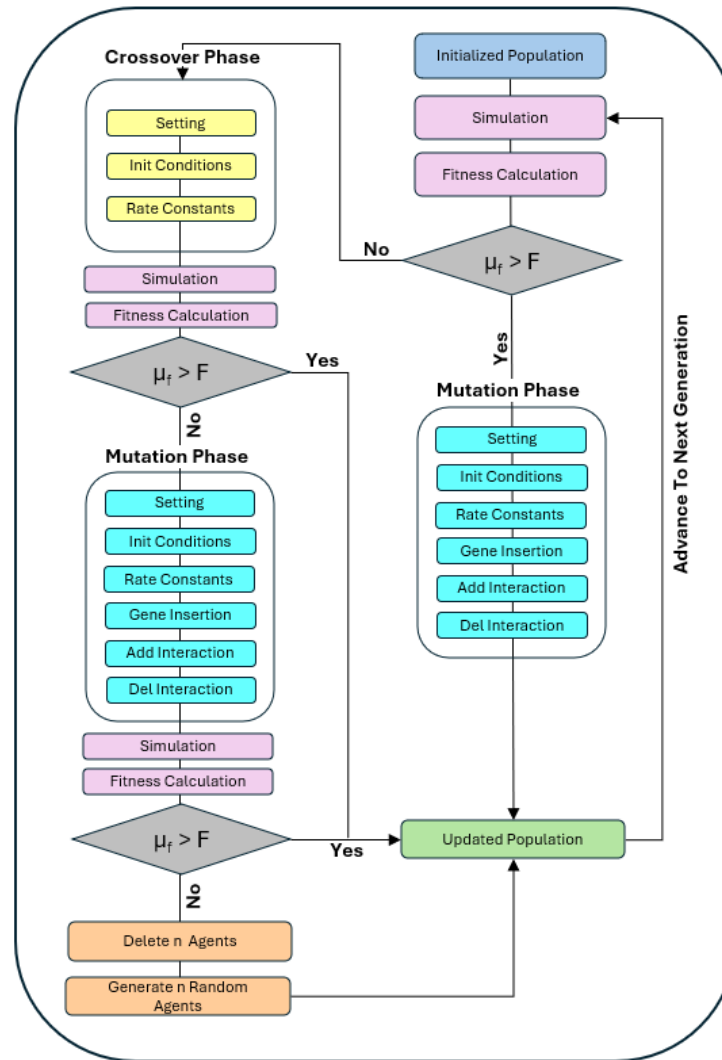
- Introduce stochastic modeling (e.g., Gillespie algorithm) to capture biological noise

Computational Bottleneck

- Rewrite in C++ for faster, large-scale simulations

Thank You !





Gradient w.r.t. constructed pattern:

$$\frac{\partial L_t}{\partial \hat{y}} = \alpha \cdot \frac{\partial L_{\text{mse}}}{\partial \hat{y}} - \beta \cdot \frac{\partial L_{\text{ssim}}}{\partial \hat{y}}$$

Gradient w.r.t. parameter:

$$\frac{\partial L_t}{\partial \theta} = \frac{\partial L_t}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial C} \cdot \frac{\partial C}{\partial \theta}$$

First moment estimate (m):

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \nabla_{\theta} L_t$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) (\nabla_{\theta} L_t)^2$$

Second moment estimate (v):

Bias-corrected estimates ($\hat{m}$, $\hat{v}$):

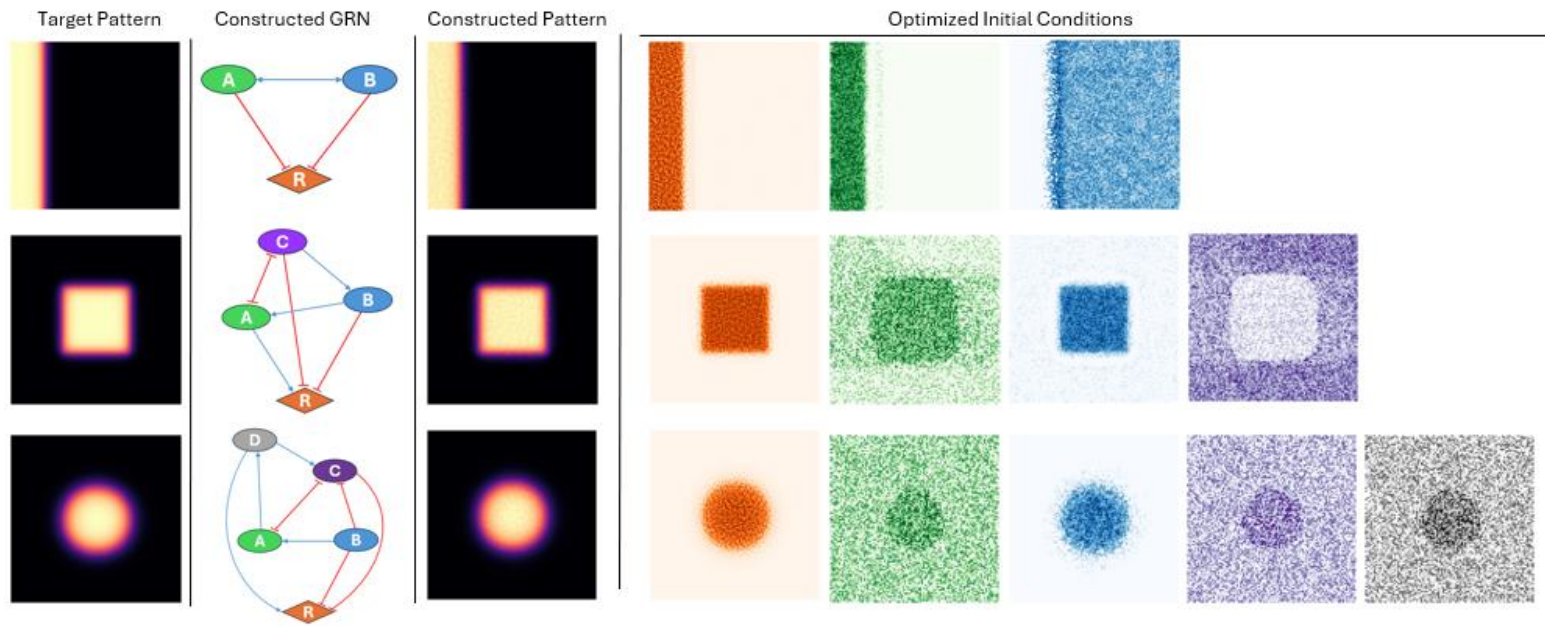
$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

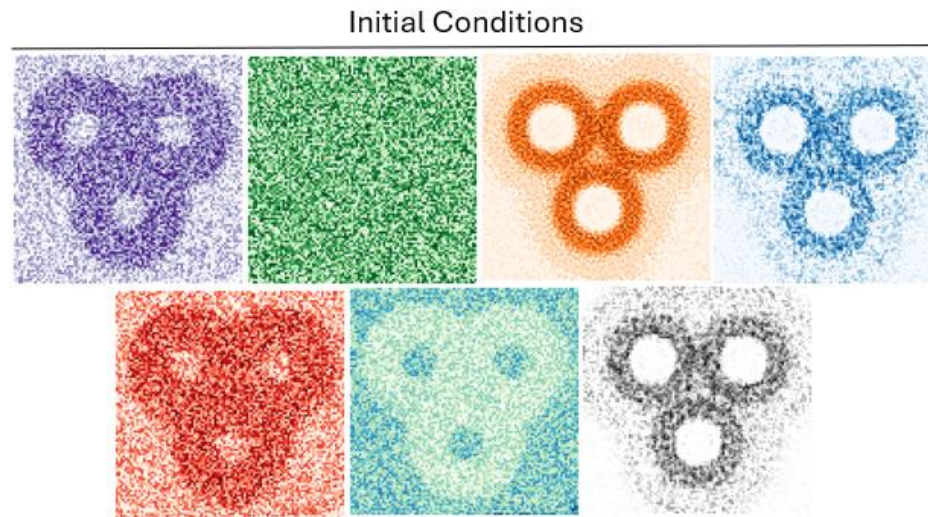
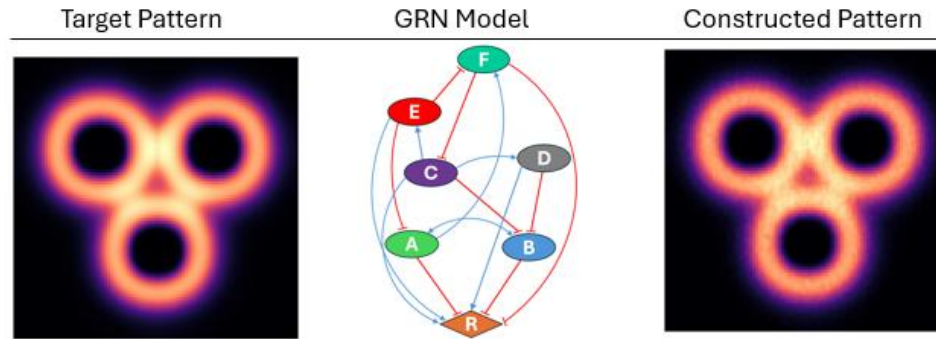
Updated parameter (θ'):

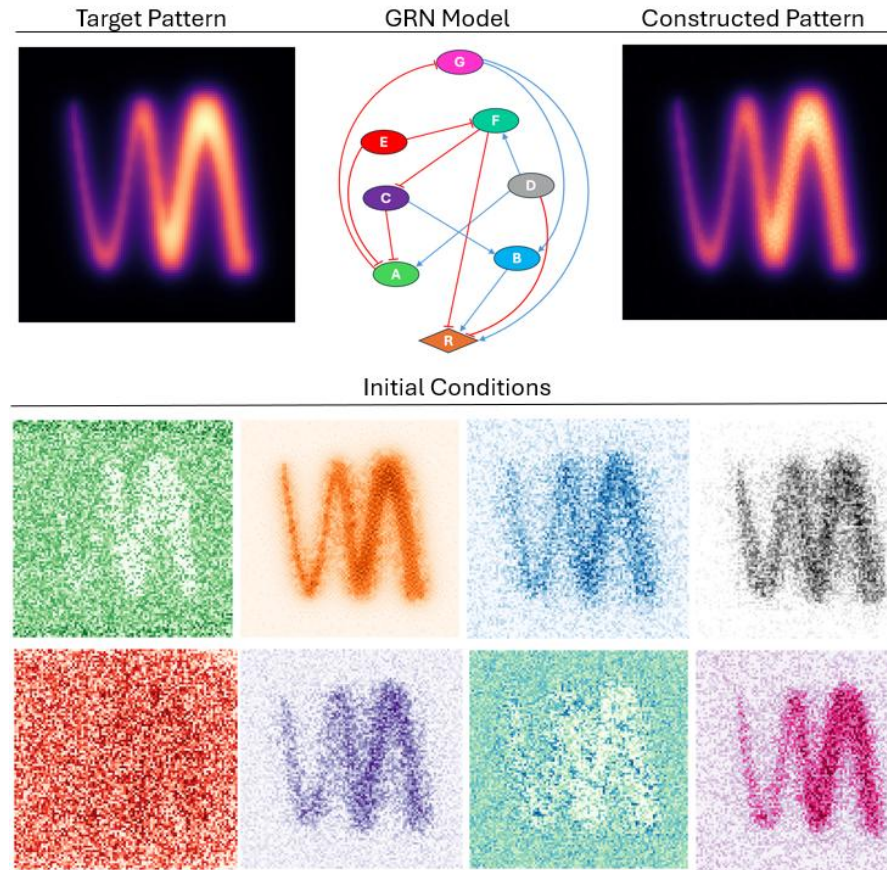
$$\theta' = \theta - \alpha \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

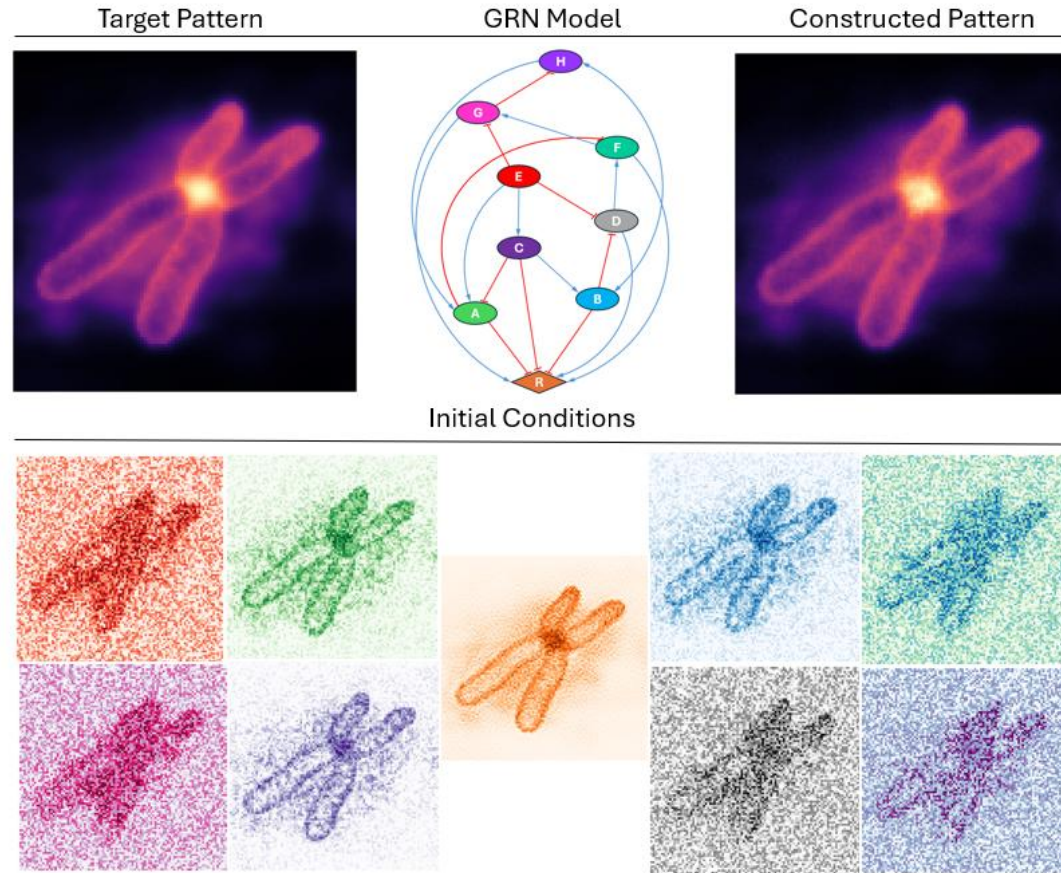
Model	Pattern	Genes	Interactions	Trainable Parameters
1	Gradient	3	4	30,021
2	Rectangle	4	7	40,033
3	Circle	5	8	50,039
4	Infinity	6	10	60,048
5	Rings	7	15	70,066
6	M-Shape	8	13	80,063
7	Chromosome	9	18	90,081
8	DNA Strand	10	21	100,093
9	Biohazard	12	25	120,111

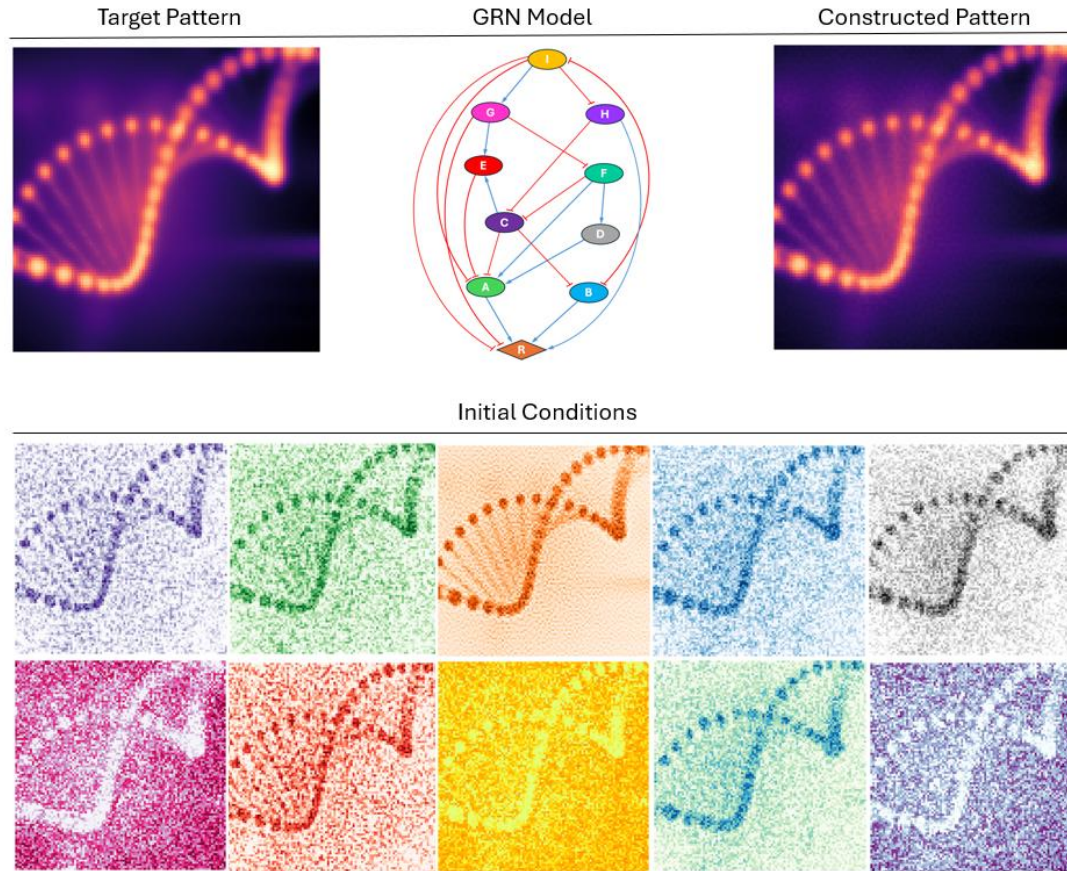
$$\begin{aligned}
\frac{\partial R}{\partial t} &= 0.936 \cdot \rho(x, y) - 0.270 \cdot R + \frac{0.894 \cdot A^{0.100}}{0.573^{0.100} + A^{0.100}} + \frac{0.100 \cdot B^{0.692}}{0.648^{0.692} + B^{0.692}} \\
&\quad - \frac{0.608 \cdot E^{0.732}}{0.312^{0.732} + E^{0.732}} + 0.999 \cdot \phi(x, y) \\
\frac{\partial A}{\partial t} &= 0.999 \cdot \rho(x, y) - 0.790 \cdot A + \frac{0.450 \cdot C^{0.375}}{0.699^{0.375} + C^{0.375}} - \frac{0.236 \cdot D^{0.100}}{0.648^{0.100} + D^{0.100}} \\
&\quad + 0.579 \cdot \phi(x, y) \\
\frac{\partial B}{\partial t} &= 0.325 \cdot \rho(x, y) - 0.210 \cdot B + \frac{0.756 \cdot D^{0.139}}{0.444^{0.139} + D^{0.139}} - \frac{0.367 \cdot A^{0.395}}{0.232^{0.395} + A^{0.395}} \\
&\quad - \frac{0.745 \cdot E^{0.928}}{0.259^{0.928} + E^{0.928}} + 0.151 \cdot \phi(x, y) \\
\frac{\partial C}{\partial t} &= 0.195 \cdot \rho(x, y) - 0.209 \cdot C + 0.070 \cdot \phi(x, y) \\
\frac{\partial D}{\partial t} &= 0.999 \cdot \rho(x, y) - 0.183 \cdot D + 0.733 \cdot \phi(x, y) \\
\frac{\partial E}{\partial t} &= 0.681 \cdot \rho(x, y) - 0.853 \cdot E - \frac{0.390 \cdot A^{0.101}}{0.999^{0.101} + A^{0.101}} - \frac{0.450 \cdot C^{0.375}}{0.439^{0.375} + C^{0.375}} \\
&\quad + 0.075 \cdot \phi(x, y)
\end{aligned}$$



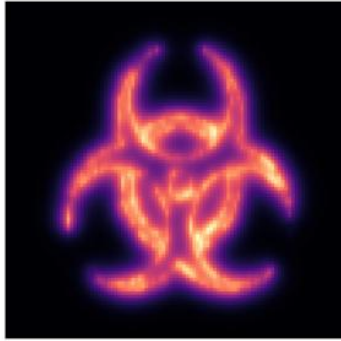




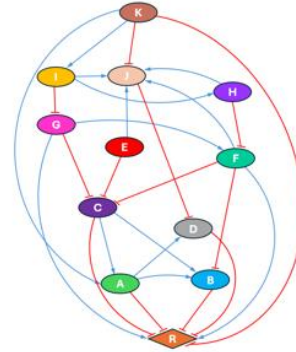




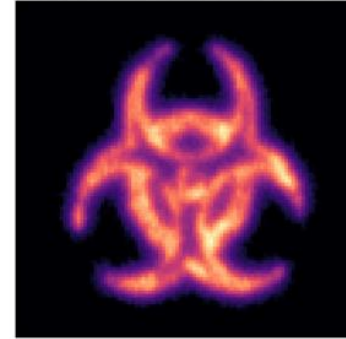
Target Pattern



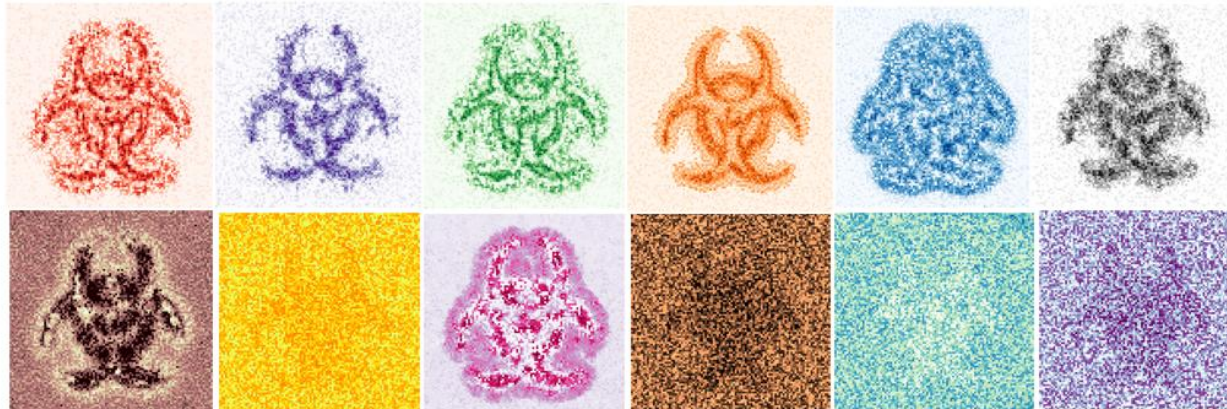
GRN Model



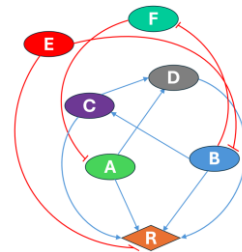
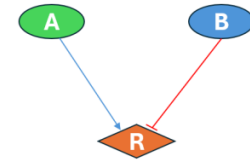
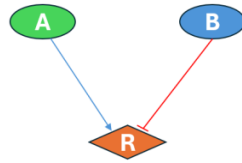
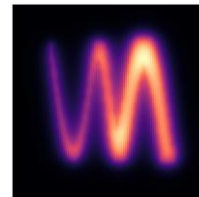
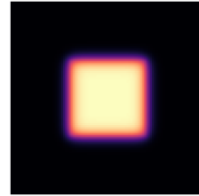
Constructed Pattern



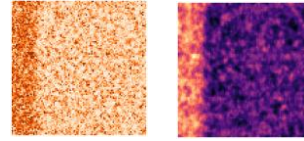
Initial Conditions



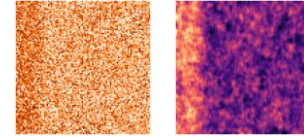
- downscaling the target pattern using third-order interpolation
- Reducing the search domain
- Reducing the computational cost



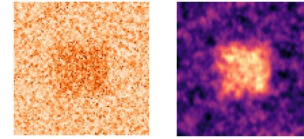
With Downscaling



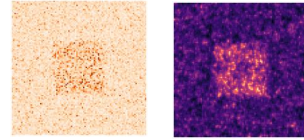
Without Downscaling



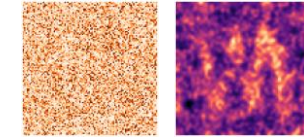
With Downscaling



Without Downscaling



With Downscaling



Without Downscaling

